

AI-Assisted Curbside Code Enforcement

A Privacy-First Municipal Fleet Imaging Framework

Prepared For: Municipal Governments & Public Works Departments

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Revision History

Version	Date	Description	Author
1.0	2026-02-19	Initial professional white paper release	GeeksDoByte / Rayen Kamta

Executive Summary

Municipal code enforcement departments face increasing pressure to ensure compliance while operating with limited staffing and reactive workflows. This white paper proposes an AI-assisted, human-reviewed system leveraging existing garbage truck routes to detect visible curbside construction indicators that may require permitting.

The system is designed as a decision-support platform, not automated enforcement. All detections require human review prior to any municipal action.

System Architecture Overview

The proposed system consists of the following pipeline:

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graph TD; A[Garbage Truck Cameras] --> B[Secure Upload (LTE / Depot Wi-Fi)]; B --> C[Privacy Redaction (Faces & Plates Blurred)]; C --> D[AI Detection Pipeline (Dumpsters, Debris, Materials)]; D --> E[Event Aggregation + GPS Mapping]; E --> F[Human Review Dashboard]; F --> G[Optional Case Creation];
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Garbage Truck Cameras
Secure Upload (LTE / Depot Wi-Fi)
Privacy Redaction (Faces & Plates Blurred)
AI Detection Pipeline (Dumpsters, Debris, Materials)
Event Aggregation + GPS Mapping
Human Review Dashboard
Optional Case Creation

Privacy & Governance Framework

The system is designed under a privacy-by-design architecture. Imaging is restricted to public rights-of-way. Automatic redaction is applied prior to review. Raw video retention is limited to 7–14 days. Flagged events are retained for defined enforcement lifecycle periods.

Key safeguards include:

- No facial recognition
- No automated citation issuance
- Human-in-the-loop review required
- Role-based access controls
- Full audit logging

Operational Benefits

- Improved inspection targeting efficiency
- Reduced complaint-driven bias
- Increased permit compliance visibility
- Enhanced data transparency
- Scalable infrastructure leveraging existing municipal routes

By leveraging existing fleet infrastructure, municipalities can modernize enforcement without expanding field staffing or engaging in invasive surveillance.